Columbus Micromobility Data

System Architecture Diagram

Overview

A GitHub Actions-driven data pipeline: **collect** → **process** → **publish**

1. DATA SOURCES (Public APIs)

All data feeds are public and require no authentication.

GBFS Spin	feeds.spin.app	Vehicle positions, battery, availability
GBFS Veo	gbfs.veo.dev	Vehicle positions, battery, availability
311 Complaints	Columbus GIS REST API	Parking & safety reports (30-day rolling window)
Static Data	Committed once	Ride Hubs, policy zones, municipality boundaries

2. GITHUB ACTIONS AUTOMATION

Three scheduled workflows that run on GitHub's free tier with no external servers.

pull-gbfs-data.yml	8am + 8pm UTC	Runs notebook, triggers fleet-export	CSV snapshots
fleet-export.yml	4am UTC + triggered	Fetches live GBFS, rebuilds dashboard	JSON + HTML
pull-311-data.yml	Every hour	Fetches 311 API, rebuilds dashboard	JSON + HTML

3. DATA PROCESSING & STORAGE

Timestamped Snapshots

• **/snapshots/** — CSV (GBFS) and JSON (311) files

• 90-day retention window

• Examples: columbus_scooters_YYYYMMDDTHHMMSSZ.csv, 311_requests_YYYYMMDDTHHMMSSZ.json

Consolidated Data Files

• **data-gbfs.json** — Latest normalized fleet data

• **data-gbfs-observations.json** — Snapshot-to-snapshot changes

• **data-311.json** — Latest complaint data

Repository Storage

• All data committed to main branch

• Full git audit trail available

• Public domain license

4. PUBLIC OUTPUTS (GitHub Pages)

Live Dashboard

URL: https://steveneedham.github.io/columbus-micromobility-data/

Updates: Hourly (311) + 2× daily (GBFS)

Features: Interactive map, multiple layers, compliance tabs, vehicle filters, municipality views

Scooter Challenge App

URL: /scooter-challenge/

Type: React single-page app for cost comparison (Spin vs. Veo)

Features: Live GBFS, cost projection, vehicle availability by location, skill export

Neighborhood Sites

URL: /micro-sites/

Area-specific dashboards for Downtown, Short North, OSU, and other neighborhoods

Raw Data Access

• CSV: raw.githubusercontent.com/.../columbus_scooters_latest.csv

• JSON: raw.githubusercontent.com/.../311_requests_latest.json

• License: Public domain

5. USERS & STAKEHOLDERS

City Planners	Compliance tracking, hotspot analysis
Researchers	Policy analysis, historical data
Individual Users	Cost comparison, vehicle availability
Community Advocates	Parking concerns, safety insights

Key Characteristics

✓ Fully automated — No manual data pulls
✓ Public data only — No credentials required
✓ Version controlled — Audit trail via git
✓ Low overhead — Runs on GitHub's free tier
✓ Two main outputs — Dashboard + cost comparison app
✓ Transparent — All code and data open to public review